

INTERNAL REFERENCE DOCUMENT

CGM source matrix

What time resolution the user actually sees inside Glev — depending on sensor, the source app on their phone, and the chosen Glev source.

Quick lookup

SENSOR	SOURCE APP ON PHONE	GLEV SOURCE	POINTS / HOUR	LATENCY	GAP RISK
Libre 2	LibreLink	LLU	4 / 15 min	2-4 min	medium — hypos between ticks invisible
Libre 2	xDrip4iOS / iAPS / Loop	Apple Health	60 / 1 min	30 s - 2 min	low (as long as phone is in BLE range)
Libre 2	xDrip4iOS / iAPS / Loop	Nightscout	60 / 1 min	30 s - 1 min	low
Libre 3	LibreLink (DACH)	LLU	4 / 15 min	2-4 min	medium — Abbott aggregates for Share API
Libre 3	LibreLink (US/UK, older)	Apple Health	60 / 1 min	30 s - 2 min	low
Libre 3	xDrip4iOS / Juggluco	AH or NS	60 / 1 min	30 s - 2 min	low
Dexcom G6	Dexcom app	Apple Health	12 / 5 min	5-7 min	low (hardware limit)
Dexcom G7	Dexcom app	Apple Health	12 / 5 min	5-7 min	low
Dexcom G6/G7	Loop / Loop Follow / xDrip	Nightscout	12 / 5 min	1-3 min	low

Important: Glev refreshes its UI every 60 s (with a 30 s client cache). Even with 1-minute data sources, the user sees at most one new point per minute — but the source decides how much real resolution sits behind it.

What the three Glev sources actually mean technically

1. LLU (LibreLinkUp) — cloud, every 15 min

Glev calls Abbott's Share-with-caregivers API every 60 s. Abbott returns the current value (live) plus roughly the last 12 h as 15-minute aggregates. Short hypos or hypers that bottom out between two 15-min ticks are invisible to Glev — regardless of which Libre sensor. Upside: zero setup. Downside: 15-minute resolution is clinically coarse.

2. Apple Health — local, resolution = whatever the source app writes in

Glev reads from the iOS Health database. Cadence is 1:1 whatever the writing app produces. LibreLink ' Health no longer works in DACH (Abbott removed the toggle in 2023). The Dexcom app ' Health writes natively every 5 min. xDrip4iOS / iAPS / Loop / Juggluco ' Health = 1 min, because they read the sensor directly via BLE. Upside: no cloud latency. Downside: setup needed (source app, permissions, Glev iOS shell with HealthKit entitlement).

3. Nightscout — your own server, resolution = whatever the uploader app pushes

Glev polls a Nightscout endpoint every 60 s. xDrip / Loop / Juggluco ' NS = 1 min (Libre) or 5 min (Dexcom hardware limit). LibreLink ' NS is not direct, you need a middleware layer. Upside: source of truth lives with the user, platform-agnostic (iOS and Android). Downside: server hosting (free on Vercel or Railway, but 2-4 h setup).

Caveats for user communication

- DACH Libre 2 or Libre 3 users without xDrip or Nightscout are stuck on 15-minute LLU resolution. The Apple Health path is effectively dead for them. That's not Glev's fault — it's Abbott's business decision.
- Dexcom users never get 1-minute resolution, regardless of path — the sensor itself only samples every 5 min. Hardware, not software.
- G7 is not "better" than G6 for resolution — G7 is just smaller and faster on warm-up.
- Latency is not the same as resolution. An LLU curve with 15-minute points can show the latest value within 2 min (low latency), but the gaps between ticks stay empty.
- Hypo detection in Insights only counts points below 70 mg/dL that actually appear in the data stream. With 15-min LLU that means only hypos that happen to land on a tick or last at least 15 min. With 1-minute sources, basically everything.
- Manual fingersticks bypass the resolution limit — they always count and are drawn as squares on the chart. If you're on LLU and suspect a low between ticks, log a fingerstick.

Suggested helper copy for the Settings 'CGM source picker

LLU	"Resolution 15 min. Simplest setup, but short hypos can be missed — log a fingerstick if you suspect one."
Apple Health	"Resolution = your source app (e.g. xDrip = 1 min, Dexcom app = 5 min). Requires HealthKit permission in the Glev iOS app."
Nightscout	"Resolution = your uploader (xDrip = 1 min, Loop = 5 min). Requires your own Nightscout server."